Calibration Data Requirements of Machine Learning Classifiers for EEG-Based P300 Spellers: A Multi-Study Learning-Curve Analysis

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***Abstract*—P300 spellers need subject-specific calibration, and methods that shorten it are usually validated on one dataset, as if the calibration a decoder needs were portable. Prior work on the bigP3BCI archive showed that a calibration score's mapping to online accuracy does not transport across cohorts; how *much* calibration is needed, and whether that amount transports, has not been measured. In 14 bigP3BCI cohorts (224 participants, 566 records) we trained a fixed shrinkage-LDA decoder on the first k calibration characters in recording order. A cohort's budget is the smallest k at which 80% of participants decode usably: 70% character accuracy (B80; 9 cohorts with valid offline decoding) or flash-level AUC ≥ 0.75 (all 14). Budgets ranged from 2 to 30 characters (15-fold; bootstrap 95% interval 6–18-fold), and learning-curve slopes were heterogeneous ($I^2 = 0.88$). To separate transport failure from sampling noise, we compared borrowed budgets with a within-cohort split-half reference. The median budget of the other cohorts missed a cohort's own budget by a median factor of 2.5 (B80) and 3.1 (AUC), against 1.5 and 1.3 between two halves of one cohort, and was systematically too short in slow cohorts and up to 7.5 times too long in fast ones. In a secondary, post-hoc analysis, a five-participant local pilot reduced the error on the remaining participants to a factor of 1.45 (AUC; borrowed 2.5) and 1.5 (B80; borrowed 1.7). No pre-specified protocol descriptor explained the differences. Calibration length should be estimated locally from a small pilot, not copied from other datasets.**

***Index Terms*—Brain–computer interface (BCI), electroencephalography (EEG), P300 speller, machine learning, calibration, learning curves, sample efficiency, dataset shift**

I. Introduction

P300 spellers let people with severe motor impairment, including amyotrophic lateral sclerosis (ALS), communicate by attending to flashing characters [1]. Each new user must first copy-spell while labelled EEG trains a classifier. Calibration costs time and adds fatigue, and a minority of users never reach usable control at all [2], so shortening calibration is a central goal of the field.

Most work proposes a method and shows that it needs less calibration on one dataset. Self-supervised pretraining with P300 aggregation, for example, was reported to cut calibration by 60% on BCI Competition II-B [3]; transfer and semi-supervised methods make similar single-dataset claims [4]–[6]. A budget found sufficient in one setting is then implicitly treated as sufficient elsewhere, although P300 protocols differ in grid size, flash paradigm, timing, electrodes, test procedure and population.

Gorenshtein et al. showed on bigP3BCI that calibration-derived decoder discriminability predicts online accuracy within cohorts, but that the fitted mapping does not transport across them [7]. Their analysis used the full calibration block and asked whether a calibration *score* transports. The quantity that calibration-reduction methods report, and that a new site must choose, is different: *how many* calibration characters a decoder needs. In machine-learning terms this is sample efficiency under dataset shift: does a classifier's learning curve [8], and hence its required training-set size, generalise across data-generating conditions? A transport failure is also only meaningful relative to how precisely a budget can be estimated within one protocol from a finite sample of users, which earlier analyses did not quantify. Our contributions are: (i) calibration learning curves and budgets for a fixed reference decoder in 14 cohorts, with calibration truncated strictly in recording order and participant-bootstrap intervals; (ii) a transport test that compares borrowed budgets with a within-cohort split-half reference, so that sampling noise is not mistaken for non-transport; (iii) as a secondary, post-hoc analysis, a constructive alternative: a local pilot of a few participants, evaluated on held-out participants of the same cohort; and (iv) open code that reproduces every result.

II. Related Work

A. Reducing calibration

Approaches include subject-independent and self-trained CNNs [4], semi-supervised adaptation [5], self-supervised pretraining [3], and EEG foundation models pretrained on large multi-dataset corpora [9]; transfer learning for EEG has been reviewed in [6]. Each is evaluated on one or a few datasets.

B. Cross-dataset P300 analysis

MOABB benchmarks ERP pipelines at a fixed training size [10]; signal alignment improves classifier transfer across P300 datasets [11]; and decoders trained on bigP3BCI learn systematically different P300 representations in ALS and neurotypical cohorts [12]. None measures how much calibration a new cohort needs.

C. Predicting performance from calibration

Calibration-derived indices predict P300 speller accuracy within a study, but Gorenshtein et al. found the fitted mappings to be cohort-specific, with a stopping-rule proxy (time per selection) explaining 66% of the between-cohort slope variance [7]. They asked whether a calibration *score* transports; we ask whether a calibration *amount* does, how large the error is relative to within-cohort sampling noise, and what a new site should do instead.

III. Data

A. Archive

We used bigP3BCI v1.0.0 [13]: 20 visual P300 speller studies recorded in BCI2000 [14] at Duke University and East Tennessee State University, comprising 6,980 EDF+ files with a shared 16-channel montage at 256 Hz. Each session has a calibration (Train) phase without feedback and a Test phase with online feedback. Per-character flag channels record which characters each flash lit, which allows character decoding for every flash paradigm, including row–column and checkerboard layouts [15].

B. Eligibility

The unit of analysis is a participant × session × condition record with its own Train and Test files. Studies C and P were excluded because their online outcomes cannot be recovered [7]. Studies D, F, G and H were excluded because calibration and testing were recorded in different conditions. Cohorts with at least 5 participants and 4 evaluable budget levels were eligible, giving 14 cohorts (Table I).

IV. Methods

A. Preprocessing

The 16 shared channels were band-pass filtered (0.5–30 Hz, 4th-order zero-phase Butterworth) and epoched from -200 to 800 ms around each flash. Epochs were baseline-corrected (-200 to 0 ms), and the 0–800 ms post-stimulus window was decimated to 64 Hz, giving 832 features per epoch (16 channels × 52 samples). Calibration epochs exceeding 150 μV were excluded from training; test epochs were decoded as recorded. All analysis settings were fixed in a configuration file before the full run, and none was tuned on test data. EDF files were read with MNE-Python [16], filtering used SciPy [17], classification used scikit-learn [18], and statistical models used statsmodels [19].

TABLE I
ELIGIBLE COHORTS. ACC.: CHARACTER ACCURACY AT FULL CALIBRATION (OFFLINE / ONLINE). REJ.: SHARE OF CALIBRATION EPOCHS $> 150\,\mu$V. ✓: OFFLINE WITHIN 0.10 OF ONLINE.

Cohort	N	Rec.	Grid	ISI	Rej.	Acc.	Valid
A	13	39	72	125	.00	.82/.79	✓
B (ALS)	18	56	36	250	.01	.92/.90	✓
E	8	8	72	125	.00	.80/.92	
I	13	26	72	125	.23	.47/.65	
J	20	40	36	125	.35	.43/.70	
K	5	16	72	125	.61	.29/.75	
L (ALS)	11	33	36	250	.00	.86/.84	✓
M	21	42	72	125	.26	.54/.82	
N (ALS)	8	16	36	250	.21	.63/.69	✓
O	17	34	72	125	.01	.91/.88	✓
Q	36	108	72	125	.01	.83/.83	✓
R	20	80	72	188	.01	.97/.96	✓
S1	10	20	72	188	.00	1.00/1.00	✓
S2	24	48	72	188	.00	.97/.98	✓
Total	224	566					9

B. Machine-learning formulation

Each flash yields a feature vector $\mathbf{x} \in \mathbb{R}^d$, $d = 832$, and a label $y \in \{0, 1\}$ (target or non-target), giving a binary classification problem with class imbalance from about 1:4 to 1:11 (median target rate 8–20% across cohorts). Features are standardised using calibration statistics only. We use shrinkage linear discriminant analysis, the standard P300 classifier [20], which scores a flash as

$$s(\mathbf{x}) = \mathbf{w}^\top \mathbf{x} + b, \qquad \mathbf{w} = \hat{\Sigma}_\lambda^{-1}(\hat{\boldsymbol{\mu}}_1 - \hat{\boldsymbol{\mu}}_0), \tag{1}$$

where $\hat{\boldsymbol{\mu}}_c$ are class means and $\hat{\Sigma}_\lambda$ is the prior-weighted average of shrunk class covariances,

$$\hat{\Sigma}_\lambda = \sum_{c \in \{0,1\}} \pi_c \big[(1 - \lambda_c)\, \mathbf{S}_c + \lambda_c \operatorname{diag}(\mathbf{S}_c)\big], \tag{2}$$

with $\mathbf{S}_c$ the sample covariance of class c, π_c its share of training epochs, and $\lambda_c \in [0, 1]$ chosen analytically by the Ledoit–Wolf estimator [21], shrinking each class covariance towards its diagonal. Shrinkage keeps $\hat{\Sigma}_\lambda$ well conditioned when training epochs are few relative to d, the regime studied here; nothing is tuned per subject or cohort.

1) Character inference: A test character is spelled with a sequence of flashes $\mathcal{F}$; let $\mathcal{F}_c \subseteq \mathcal{F}$ be the flashes that lit candidate character c. The decoded character is

$$\hat{c} = \arg\max_c \; \frac{1}{|\mathcal{F}_c|} \sum_{f \in \mathcal{F}_c} s(\mathbf{x}_f), \tag{3}$$

using all flashes the original protocol presented. The mean (rather than sum) makes the rule valid for checkerboard paradigms, where characters are lit unequally often.

2) Learning curves: For each record and each training-set size $k \in \{1, 2, 3, 4, 6, 8, 10, 12, 15, 20, 25, 30, \ldots, \text{full}\}$ calibration characters, the classifier is trained only on the first k characters in recording order (72–180 epochs per character, median 120), so smaller training sets are strict subsets of larger ones and evaluation never uses calibration data. This yields

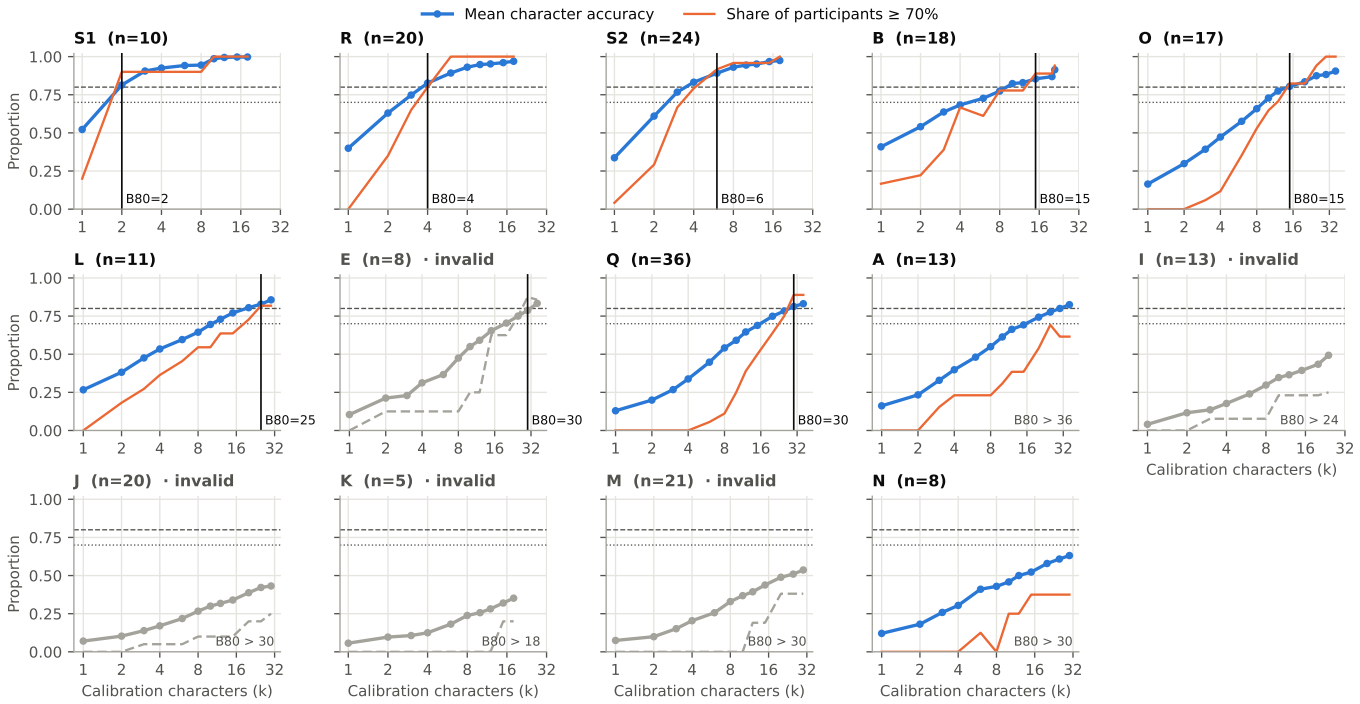


Fig. 1. Calibration learning curves per cohort: mean character accuracy (blue) and share of participants at $\geq 70\%$ (orange) versus calibration characters k. Vertical line: B80. Grey panels ("invalid"): offline decoding not valid (dynamic-stopping tests).

6,488 trained models (plus 562 shuffled-label controls) across 562 records, i.e. an empirical learning curve per record [8]; in 4 records (cohorts I and J) too few calibration epochs survived artefact rejection to train a classifier.

3) Controls and verification: A label-permutation control (shuffled calibration labels) gave chance-level accuracy in every cohort (at most 0.01 above chance), and real labels exceeded it by at least 0.28. With full calibration, offline accuracy reproduced the original online accuracy in pilot sessions (0.94 vs. 0.92; 0.93 vs. 0.93). An independent re-implementation of truncation, training and decoding, written separately from the pipeline, reproduced the accuracy of randomly sampled records at all budgets exactly (maximum difference 10^{-16}), found no test epoch in any calibration set, and recomputed every budget identically.

C. Calibration budgets

Participant accuracy at k was pooled over that participant's records. A budget k was evaluable when at least 80% of a cohort's participants had data at k. B80 is the smallest evaluable k at which at least 80% of participants reach 70% character accuracy, right-censored if none qualifies. In five cohorts (E, I, J, K, M) offline accuracy fell more than 0.10 below online accuracy. We attribute this to dynamic stopping [22]: online sequences ended when the online classifier was confident, which disadvantages any other decoder (N and O also stopped early but stayed within tolerance). B80 is therefore interpreted in the 9 cohorts with valid decoding; including all 14 leaves its range unchanged (2–30 characters), because E reaches B80 at 30 and I, J, K, M are censored. The co-primary AUC budget applies the same rule to flash-level AUC ≥ 0.75 (mean over a participant's records), which does not depend on test length, in all 14 cohorts. Both the validity rule and the AUC budget were added after inspecting full-calibration accuracy and before any budget was computed. Intervals came from 2,000 participant-cluster bootstrap resamples.

D. Statistical analysis

Stage 1 fitted, per cohort, a binomial GLM of correct characters on $\log k$ (or OLS of AUC on $\log k$), centred at $k = 4$, with participant-clustered standard errors. Stage 2 pooled cohort slopes and intercepts with DerSimonian–Laird random effects [23], reporting τ, I^2 [24], a Hartung–Knapp confidence interval [25] and a prediction interval. Seven pre-specified protocol descriptors (seconds per calibration character, test flashes per character, grid size, inter-stimulus interval, row–column share, characters lit per flash, ALS cohort) entered one at a time in a Knapp–Hartung meta-regression (scale factor floored at 1) of slopes and, separately, of intercepts. Artefact rejection rate was added afterwards as an exploratory descriptor, motivated by the high rejection rates in the cohorts that never reached a budget; Holm correction [26] was applied across all eight. The stage-2 code recovered a known moderator effect, and no spurious one, on simulated data.

E. Transportability and local pilot

Each cohort was withheld in turn and its budget predicted from the remaining cohorts' pooled participants (primary) or

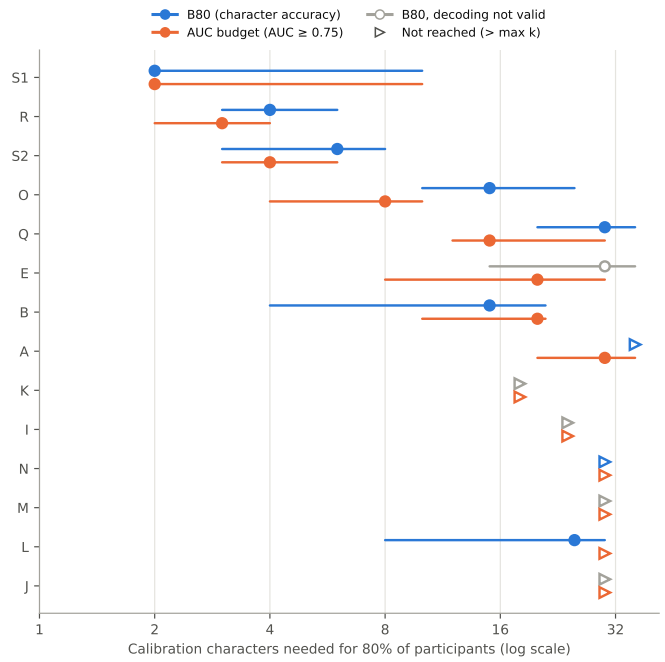


Fig. 2. Calibration budget per cohort: B80 (blue) and AUC budget (orange) with participant-bootstrap 95% CIs. Open triangles: not reached. Grey: B80 where decoding was not valid.

from the median of their budgets (borrowed budget). Errors are $|\log_2(\hat{B}/B)|$, so a factor-of-two miss counts as 1 at any budget size. Two comparisons were added after an internal audit, before their results were seen. The *within-cohort reference* splits a cohort's participants at random into halves (500 splits) and compares the halves' budgets, which is the error expected from sampling users alone under one protocol. The *local pilot* estimates the budget from $n \in \{3, 5, 8\}$ randomly drawn participants of the same cohort (500 draws; full calibration if the pilot never meets the criterion). Pilot and borrowed budgets are scored on the cohort's remaining participants, by their error relative to those participants' budget and by whether at least 80% of them are usable at the recommended length. Each cohort's median counts once.

V. Results

A. Learning curves differ between cohorts

Accuracy rose with k in every cohort, but at different rates and to different levels (Fig. 1). In valid cohorts the pooled slope was 1.06 log-odds per $\log k$ (95% CI 0.83–1.28), with between-cohort heterogeneity ($\tau = 0.20$, $I^2 = 0.88$, $p < 0.001$); the 95% prediction interval for a new cohort was 0.55–1.56. Intercepts varied more ($\tau = 0.98$, $I^2 = 0.96$). On AUC across all 14 cohorts the pooled slope was 0.067 per $\log k$ ($I^2 = 0.84$).

B. Calibration budgets vary at least sixfold

B80 ranged from 2 characters (S1) to 30 (Q) among the 7 valid cohorts that reached it, a 15-fold spread; A and N did not reach it. The AUC budget gave the same picture across all 14 cohorts, ranging from 2 to 30 characters (15-fold) among the 8 that reached it, and ranked cohorts similarly (Fig. 2). In time, the AUC budget ranged from 45 s (S1) to 900 s (B). Four dynamic-stopping cohorts (I, J, K, M) reached neither budget, even at an AUC threshold of 0.70. Because the max/min ratio is an extreme-value statistic from cohorts of 10–36 participants, we bootstrapped it jointly over participants: its 95% interval was 6–18-fold for B80 and 7.5–15-fold for the AUC budget, and it was at least fivefold in over 99% of resamples. Omitting S1 alone reduces the B80 ratio to 7.5. Censored cohorts are excluded, so the true spread can only be larger.

C. Protocol descriptors do not explain the differences

None of the pre-specified protocol descriptors was associated with learning-curve slope after Holm correction (Fig. 3), nor with the cohort intercept, i.e. the level at $k = 4$ (minimum Holm $p = 0.15$), so H3 was not supported. With 9–14 cohorts these tests have low power and cannot exclude moderate effects. In valid cohorts, grid size, characters lit per flash and ALS status each explained about 38% of slope variance ($p \approx 0.03$, Holm $p = 0.24$); they are collinear, since the ALS cohorts also used 36-character grids. In the exploratory analysis, artefact rejection rate was associated with the AUC slope ($\beta = -0.012$ per SD, $p = 0.002$, Holm $p = 0.016$; 0.031 across both outcome families). The association survived omitting any single cohort (all $p \leq 0.008$) but halved without the four high-artefact, early-stopping cohorts I, J, K, M ($\beta = -0.007$, $p = 0.054$), and it is partly mechanical, since rejected epochs leave fewer training epochs per calibration character. It is a hypothesis, not an explanation.

D. Borrowed budgets do not transport

Pooling all other cohorts' participants produced a budget at which 80% of them were usable only once (withholding Q, B80): no single calibration length served 80% of users across studies. A naive error criterion is uninformative here, because two random halves of the *same* cohort already disagree by at least two characters in 65% (B80) and 55% (AUC) of splits. We therefore compared log-scale errors with this within-cohort reference (Fig. 4). Budgets from two halves of a cohort differed by a median factor of 1.5 (B80) and 1.3 (AUC). The median budget of the other cohorts missed a withheld cohort's own budget by a median factor of 2.5 (B80, 7 cohorts with a finite budget) and 3.1 (AUC, 8 cohorts), and by at least the lower bounds shown for censored cohorts. The error was systematic: in slow cohorts (A, B, L, N, O, Q) fewer than 30% of held-out evaluations had 80% of participants usable at the borrowed budget, whereas in fast cohorts (R, S1, S2) it was 2.5–7.5 times longer than needed. In the secondary pilot analysis, a pilot of five participants from the same cohort removed most of this bias: its median error on the remaining participants was a factor of 1.45 for the AUC budget (borrowed 2.5) and 1.5 for B80 (borrowed 1.7), close to the within-cohort reference; with eight participants, 1.33 and 1.5. Neither rule reliably met the 80% target on new participants (pilot 34–38%,

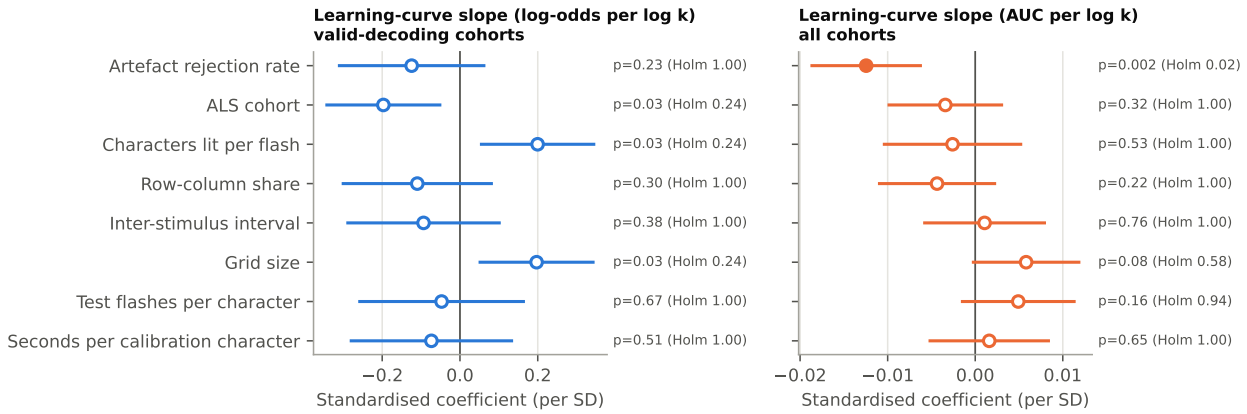


Fig. 3. Standardised meta-regression coefficients (±95% CI) of cohort learning-curve slope on each descriptor. Filled: significant after Holm correction.

TABLE II
BUDGET ACROSS ROBUSTNESS VARIANTS: COHORTS REACHED, RANGE (CHARACTERS), MAX/MIN RATIO.

Variant	B80	AUC budget
Primary	7/9, 2–30, 15×	8/14, 2–30, 15×
First session only	7/9, 2–36, 18×	7/14, 2–30, 15×
≤20% artefacts	8/9, 2–30, 15×	9/14, 2–30, 15×
Lower threshold[a]	8/9, 2–25, 12.5×	9/14, 2–20, 10×
Higher threshold[b]	5/9, 2–30, 15×	5/14, 3–36, 12×

[a] accuracy 0.60 / AUC 0.70. [b] accuracy 0.80 / AUC 0.80.

borrowed 33–40% of evaluations), because users within a cohort also differ: a cohort-level budget fixes the bias between sites, not the spread between users.

E. Robustness

The spread persisted in every variant (Table II). Removing high-artefact records let cohort N reach both budgets (B80 = 20, AUC budget = 30), consistent with the exploratory moderator result.

VI. Discussion

A. Principal findings

With one fixed decoder and rule, the calibration at which 80% of users decoded usably ranged from 2 to 30 characters (under a minute to 15 minutes). A budget borrowed from other studies was wrong by a factor of about 2.5–3, twice the within-study sampling error, and systematically so; a five-person local pilot came close to the within-study level.

B. Implications

For method papers, a claim such as "60% less calibration" is relative to a baseline that itself differs severalfold between datasets; the same absolute saving can be decisive in a slow cohort and irrelevant in a fast one. Calibration-reduction methods should therefore be evaluated on several cohorts with per-cohort budgets. For deployment, calibration lengths copied from the literature over-calibrate users in fast settings and under-calibrate them in slow ones. A site can instead calibrate its first few users fully and set the budget from their learning curves; because users within a site still

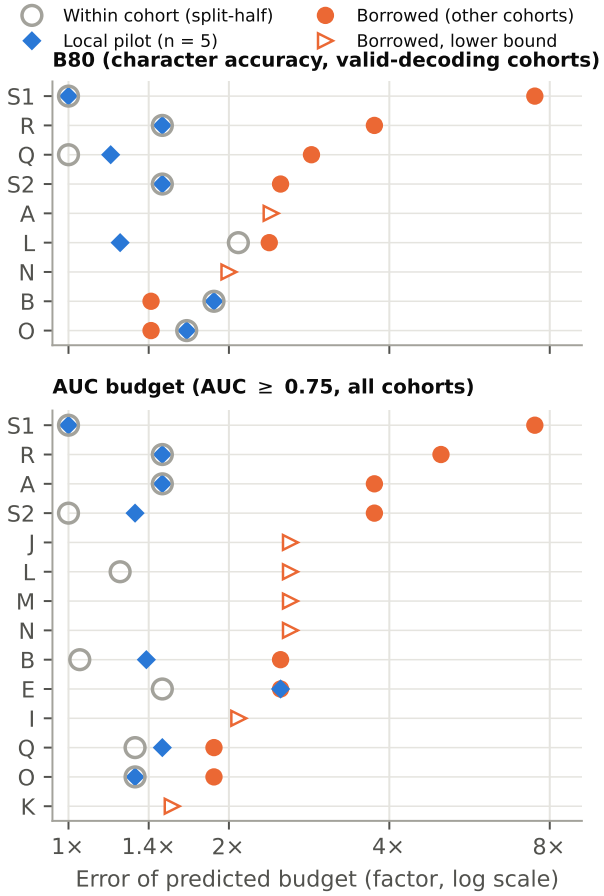


Fig. 4. Error of each cohort's predicted budget as a factor of its own budget (log scale). Grey: two random halves of the cohort (median of 500 splits). Blue: five-participant pilot, scored on the remaining participants (median of 500 draws). Orange: median budget of the other cohorts; open triangles: lower bounds (budget not reached). Missing markers: budget not reached in pilot or held-out half.

differ, per-user adaptive stopping of calibration remains the natural complement. Gorenshtein et al. found that a stopping-rule proxy explained much of the between-cohort variation in their score–accuracy mapping [7]; here test length did not explain differences in calibration learning, suggesting that the value of a calibration score and the amount of calibration needed vary for partly different reasons.

C. Limitations

The analysis is offline with a fixed decoder rather than the original online classifier, whose BCI2000 parameters are not public, and five dynamic-stopping cohorts could not be decoded validly offline. Protocol, laboratory, hardware and population are nested within cohorts, so associations are not causal and ALS effects cannot be separated from grid size or timing. Participant identities are anonymised: sex and birth year are available for 48 of 327 participants, and their agreement suggests that some participants of K may also appear in M (one also in I); excluding K leaves AUC heterogeneity unchanged ($I^2 = 0.85$), but overlap among the other cohorts, including the ALS cohorts B and N, could not be checked. The pilot analysis draws pilot and evaluation participants from the same study, so a prospective test at a new site is still needed. Finally, we tested one decoder; deep, transfer-learned or foundation-model decoders [6], [9] may lower budgets, and whether they remove between-cohort heterogeneity is left to future work.

VII. Conclusion

Across 14 cohorts and 224 participants, the calibration budget for a fixed P300 decoder varied at least sixfold (point estimate 15-fold), and a budget borrowed from other studies missed by a factor of about 2.5–3, twice the within-study sampling error. A pilot of five local participants reduced the error to near that level. Calibration-reduction studies should report per-cohort budgets, and deployments should estimate calibration length locally rather than copy it. Code that reproduces all results from bigP3BCI v1.0.0, including the independent verification, is available at https://github.com/DevanshSati18/p300-calibration-budget.

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